CAPSTONE PROJECT GUIDELINES
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

FOR (ITS410)

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(January 2024)

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Table of Contents

I. Introduction.....

II. Capstone / Research Project Agenda.....

III. Suggested Areas of Capstone / Research Project.....

Capstone / Research Project Categories

Software Development

Multimedia Systems

IV. Pre-requisites.....

V. Guidelines.....

Software (Capstone) Project 1……...
Software (Capstone) Project 2

VI. Grading System.....

Proposal Stage (Software (Capstone) Project 1)Verdicts

System Oral Defense (Software (Capstone) Project 2)Verdicts

VII. Capstone / Research Project Duration.....

Pre-proposal Stage

Proposal Stage

Oral Defense Stage

VIII. Capstone / Research Project Team.....

Regrouping Policy

IX. Capstone / Research Project Adviser.....

Responsibilities as the Subject Teacher

X. List of Deliverables/Activities

Appendix A. Project Team Assignments Form

Appendix B. Pre-Proposal Statement Template

Appendix C. Project Working Title Form

Appendix D. Progress Monitoring Report Form

Appendix E. Capstone / Research Project Manuscript Outline

Appendix F. Capstone / Research Project Hearing Notice Form

Appendix G. Grammarian's Certificate Template

Appendix H. Approval Sheet Template

Appendix I. Title Page (Proposal)

Appendix J. Title Page (Oral Defense)

Appendix K. Acceptance Sheet

I. Introduction

The **Bachelor of Science in Information Technology (BSIT)** program is designed to produce graduates equipped with the necessary skills to become competent IT professionals. The program focuses on core competencies such as software installation, systems operation, maintenance, network administration, and the development and deployment of information systems (Commission on Higher Education [CHED], 2015).

As stipulated in **CHED Memorandum Order No. 25, s. 2015**, BSIT students are required to complete a **Capstone Project** as a major course component. This project must be in the form of a **systems application** or an **enterprise resource plan (ERP)** that addresses a specific business or community need. It is an integrative course where students are expected to apply theories, methodologies, and technical skills they have acquired throughout their academic training (CHED, 2015).

The Capstone Project serves as a culminating academic and professional experience. It offers students the opportunity to showcase their ability to define problems, design and develop effective IT solutions, and manage full-cycle system implementation. Furthermore, it challenges students to demonstrate innovation, initiative, and adherence to professional and industry standards (Ala, 2020).

More than a compliance requirement, the Capstone Project is a means of evaluating a student's readiness to graduate and transition into professional practice. It ensures that students are not only technically capable but also prepared to contribute meaningfully to the information technology sector (Fabito et al., 2021).

The Capstone Project has a number of educational objectives. Although each Research / Capstone Project is different and the relative emphasis will vary, the subject will involve students in:

- Bringing together and integrating knowledge and skills in the course as a whole;
- Reinforcing and developing competencies that have not been sufficiently emphasized in the fundamental subjects;
- Defining a substantial engineering study or design task and carrying it to completion within a specified time and to a professional standard;
- Completing a comprehensive written and bound report that places the Research / Capstone Project in context, defines its objectives, and describes the work done with the resulting conclusions or recommendations;
- Demonstrating initiative and creativity, taking pride in the achievement of a difficult task.

Through this course, students are prepared in their respective careers.

I. Capstone / Research Project Agenda

The Capstone and Research Project Agenda of the **Information Technology Area** at **La Consolacion College Bacolod** is designed to support the development of innovative, socially responsible, and globally relevant technological solutions. It is anchored on the institution's vision of forming IT professionals who are competent, compassionate, and committed to nation-building. The following thrusts define the agenda:

1. Business and Industry Innovation

Development of digital solutions that enhance productivity, efficiency, and competitiveness in business and industrial operations, especially among micro, small, and medium enterprises (MSMEs).

2. Industry-Driven Research

Engagement with real-world industry problems through applied research, systems development, and technology integration tailored to the needs of various sectors including education, healthcare, retail, logistics, and agriculture.

3. Socio-Economic Development

Implementation of projects that address social needs, reduce inequality, promote digital literacy, and empower marginalized communities through inclusive and sustainable technology.

4. Global Competitiveness in the Service Sector

Design and deployment of information systems that support quality service delivery, customer satisfaction, and compliance with international standards, particularly in the BPO, tourism, education, and government sectors.

5. Good Governance and Transparency

Development of e-Government applications and monitoring systems that promote accountability, transparency, and data-driven decision-making in public service.

6. Environmental Sustainability and Climate Action

Capstone projects that contribute to environmental awareness, resource efficiency, waste reduction, and disaster risk management using smart technologies and data analytics.

7. Alignment with the United Nations Sustainable Development Goals (SDGs)

Capstone initiatives shall intentionally align with relevant SDGs, particularly:

- **SDG 4** – Quality Education
- **SDG 8** – Decent Work and Economic Growth
- **SDG 9** – Industry, Innovation, and Infrastructure
- **SDG 11** – Sustainable Cities and Communities
- **SDG 13** – Climate Action
- **SDG 16** – Peace, Justice, and Strong Institutions

It is expected that the students' capstone projects / researches shall be in parallel with the already laid down priority areas of the College, Commission on Higher Education (CHED,

PSITE-NIR(Philippine Society of IT Educators-Negros Island Region) and aligned with the national research roadmap.

II.Suggested Areas of Capstone / Research Project

Capstone / Research Project Categories

The Research or Capstone Project must be innovative, original, and relevant to a real-world organization or community of similar nature or scope. Projects must not duplicate previous works of former students and must not be based on off-the-shelf software or templates (e.g., Microsoft Access sample projects, free website builders, or existing platforms without major customization).

Capstone Projects must demonstrate custom development, system integration, and applied IT solutions. The proposed system may fall under, but is not limited to, the following categories:

1. Software Development

- Software Customization for Business or Institutional Use
- Information Systems Development (e.g., enrollment, human resource, inventory, point-of-sale systems)
- Web Application Development
- Mobile Computing Applications (Android/iOS)
- Desktop-Based Systems with Local Area Network Support
- Cloud-Based Systems with Remote Access
- Platform or Framework Development
- Data-Driven Systems with Analytics and Visualization
- System Integration with Third-Party APIs

2. Multimedia Systems

- Game Development (educational, simulation, interactive storytelling)
- E-Learning Management Systems (LMS)
- Interactive Multimedia Applications
- Information Kiosks
- Virtual Exhibits or Interactive Campus Maps
- Augmented Reality (AR) and Virtual Reality (VR) Projects
- Multimedia-Based Instructional Tools
- 3D Animated Interfaces or Tutorials

3. Emerging Technologies

- Internet of Things (IoT)-Based Applications
- Smart Home or Smart City Solutions
- Wearable Device Applications
- Blockchain-Based Transaction or Record Systems
- Artificial Intelligence (AI) or Machine Learning (ML) Applications
- Chatbots for Education, Business, or Government Use
- Computer Vision or Facial Recognition Systems

- Robotics and Automation Systems

4. Network and Infrastructure Solutions

- Network Design and Simulation Projects
- Campus-Wide Network Monitoring Tools
- Cybersecurity Tools or Firewalls
- GSM/SMS-Based Alert or Notification Systems
- VPN or Secure Communication Platforms
- Intrusion Detection Systems
- Remote Infrastructure Management Systems

5. Data-Driven and Business Intelligence Systems

- Business Intelligence Dashboards and Reporting Tools
- Predictive Analytics Applications (e.g., sales forecasting, customer behavior)
- Data Warehousing Systems
- Decision Support Systems
- CRM (Customer Relationship Management) or ERP (Enterprise Resource Planning) Systems
- Customized Financial Management Systems

6. Societal and Community-Oriented Systems

- Community-based information or incident reporting systems
- Disaster preparedness, response, and management platforms
- Digital health systems for rural or low-resource settings
- e-Government applications and public service portals
- Agricultural support and monitoring systems
- Educational tools for cultural or linguistic preservation
- Environmental data collection and visualization platforms
- Assistive technologies for persons with disabilities

Note: Students are encouraged to align their project topics with local industry needs, institutional problems, or pressing community challenges. Interdisciplinary solutions (e.g., IT in education, health, governance, agriculture, or the environment) are highly encouraged.The student must finish the following courses that prepare him/her to undergo a formal capstone project/research study

PREREQUISITES:

- The student status should be **third year standing** which means he already completed subjects like **Database Systems, Web Development**
- **ITS222 (Systems Analysis and Design)** - for Software Development steps or life cycle

II. Defense Guidelines

ITS321 (Software (Capstone) Project 1)

1) The students shall form a team of ***2 members or maximum of 3 for*** meritorious cases in which a letter of appeal is required. They then decide who plays the following roles - **Project Manager, Systems Analyst/Programmer and QA Tester/Technical Writer.** The team then submits **Project Team Assignments Form** (Appendix A) with signatures, to the Subject Teachers or to the Program Head.
2) The Proponents/Researchers of the Research / Capstone Project shall prepare 5 different possible topics/titles, and present/consult these topics to any of the IT teachers or any expert of the field.
3) The Proponents/Researchers shall make the **Pre-Proposal Statements** (*Appendix B*) of each of the selected topics/titles.
4) T**he subject teachers** and **the Program Head** shall convene and select only one of the 3 - 5 topics/titles presented. Only the approved Research / Capstone Project topics should proceed to the research proposal stage. After a topic/title is finally chosen, the team then accomplishes a **Project Working Title Form** (Appendix C) which will then indicate the name of the appropriate adviser as decided by the team of advisers together with the proponents.
5) The team shall prepare all the parts of the proposal manuscript on time with the set/agreed dates. A progress monitoring report will be provided to ***monitor the research*** project (Appendix D – Progress Monitoring Report).
6) The researchers will prepare 3 copies of the **Complete Proposal Manuscript** *(Appendix E*) for the Proposal Hearing. The **Proposal Hearing Notice Form** *(Appendix F*) from the Dean's Office should be filled out and complied. This notice and the 3 copies of complete proposal manuscript must be submitted to the Adviser.
7) The adviser select the proposal hearing panel which shall be composed of the following:
    - 1 Chairman - preferably the Program Head or a faculty with at least a master's degree.
    - 2 Members (one may be a content expert- external preferably)
8) At the end of the proposal hearing, the chair makes a synthesis and announces the panel's verdict.
9) The chairman and the adviser shall ensure that all recommendations for improvement by the proposal hearing panel are incorporated in the Proposal Manuscript.
10) The team shall prepare and provide an honorarium for the **panel of examiners**, as follows:

- **₱500** for each **regular panel member**
- **₱600** for the **Chairperson of the panel**

The **total honorarium** shall be **divided equally among the members of the project team** and must be paid **immediately after the defense proceedings**.

In addition, the team shall **allocate a separate budget for food and snacks**, to be provided equally to all panel members. This may be in the form of **in-kind provisions** (e.g., packed meals or refreshments).

11) One copy of the **Revised Proposal Manuscript** together with the **Grammarians Certificate** Appendix G) shall be routed to the Adviser, Panel members, and Chairman for the confirmation of revisions. **Acceptance Sheet** (Appendix K) may be routed too for their signatures if already amenable.
12) The **ring bound** copy containing the Acceptance Sheet and the **Final Proposal Manuscript** should be submitted to the Dean's Office.

Final Oral Defense (ITS410)

1) The team shall submit to their adviser on time (as scheduled) the **1st prototype (30% to 50%) plus 1st draft**; **2nd prototype (51% to 70%) plus 2nd draft**; and **3rd prototype (71% to 99%) plus 3rd draft** of the System/Output plus Oral Defense Manuscript, respectively.
2) If the 3rd prototype of the System/Output reaches the appropriate percentage to completion as against major modules and features and the 3rd draft of the Oral Defense Manuscript is acceptable as evaluated by the adviser, team shall then prepare and submit to the Adviser the **Oral Defense Hearing Notice**
3) The Adviser forwards the Oral Defense Hearing Notice, 1 copy of the Complete Oral Defense Manuscript to the Dean's Office.
4) The Office will then arrange the date and time of the oral defense and distribute the Manuscripts to the identified members of the oral defense panel.

   The Dean assigns qualified and competent faculty members who will constitute the Oral Defense panel. The Oral Defense panel shall be composed of the following:
   - 1 Chairman - preferably the Program Head or at a faculty with least master's degree.
   - 2 Members (one may be a content expert)
5) One of the members of the Oral Defense panel may be invited from outside the College if the study requires his/her expertise.
6) During the Oral Defense, the adviser shall be the moderator who clarifies and mediates over issues raised.
7) The Technical Writer is tasked to record all the suggestions and recommendations of the panel during the Oral Defense.
8) At the end of the Oral Defense, the chair makes a synthesis and announces the verdict.
9) The Oral Defense panel chair and the adviser shall ensure that all recommendations for improvement by the Oral Defense panel are incorporated in the final copies. This may include grammar, accuracy of language, adequacy of data, interpretation of results, etc.
10) The team shall prepare and provide for the honoraria of the panel of examiners immediately after the proceedings.
11) Approval Sheet (*Appendix H*), this time for the ITS410, is necessary prior to the final submission of the manuscript and other research transcripts.
12) The researchers must submit the **2** copies of **Oral Defense CD's**. Each of the Oral Defense CD's should contain the following:

- Developed system.
- Installation or Setup Files/Folders
- Installation and/or Users' Guide
- Complete documentation -Final Oral Defense Manuscript (word file)

III. Grading System

Proposal Stage ITS321

The Final Grade of each proponent will comprise of the following:

Average grade of the Panel Members including the Chairman	70%
Adviser of the Research / Capstone Project / Group	30%
TOTAL	100%

Verdicts

There will be three possible verdicts after the Proposal Hearing. The verdict is a unanimous decision among the three members of the Capstone Project Oral Defense panel. Once issued, it is final and irrevocable.

APPROVED. Minor revisions are necessary but they do not have to be presented in front of and checked by all panelists. 86 – 100
APPROVED WITH REVISIONS. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. 70 – 85
DISAPPROVED. The Proponents/Researchers failed to propose a researchable or scholarly Research / Capstone Project. Below 70

System Oral Defense (ITS410)

Capstone Project Output (Group Grade)	70%
Oral Examination (Individual Grade)	30%
TOTAL	100%

Verdicts

ACCEPTED WITH REVISIONS. Revisions are necessary but they do not have to be presented in front and checked by all panelists. 70 to 100
REORAL DEFENSE. Another Oral Defense session, in which all panelists must be present, is necessary to further clarify the objectives and scope of the capstone project. Student must re-apply for another Hearing Notice Form from the Center for Research if the Oral Defense is scheduled after the semester ends. 65 to 69 and upon the panel's unanimous decision
NOT ACCEPTED. The proponent failed to achieve the objectives of the research established in the proposal. The panelists' numeric grades are not anymore needed. Below 65

Defense Form

Defense Date: _____
Software Project Title: _____

PAPER		30%	
Originality	15%		_____
Quality of Project Manuscript	15%		_____
PROTOTYPE		35%	
Conceptual and Logical Design	20%		_____
Workability and Application	15%		_____
ORAL PRESENTATION		35%	
Quality of Project Presentation	20%		_____
Ability to Defend Project	15%		_____
	Total Grade		_____

Individual Grade
Proponent 1:_____ Proponent 3:_____
Proponent 2:_____ Proponent 4:_____

IV. Research / Capstone Project Duration

The BSIT Capstone Project shall be completed within two semesters (COMP401 and ITS410) with the following phases:

Pre-proposal Stage

- Course Enrolment
- Capstone Project Orientation
- Short Listing of Possible Research / Capstone Projects
- Title Critiquing
- Pre-Proposal Statement Preparation
- Pre-Proposal Hearing

Proposal Stage

- Practical Examination of the chosen Programming Language (by team) - optional
- Writing of Chapters I, II, III, and 30% (Propose System)
- Proposal Manuscript Submission
- Proposal Hearing
- Proposal Manuscript Revisions

Oral Mock Defense (Optional)

- Conducted approximately one (1) month before the Final Oral Defense, the Mock Defense is a formal rehearsal involving the same panel assigned for the final defense.
- Mock Defense Presentation (system, documentation, and slides)
- Completion Rating and Feedback Form from panel
- No formal verdict issued
- Mandatory implementation of all panel suggestions in preparation for Final Defense
- Same honorarium and dress code policies apply as in the Final Defense

Defense Stage

- Chapter IV and V
- Analysis
- Design
- Development
- Testing
- Prototype Submission (3 Loops) 1st loop- 30 50, 2nd loop- 51 to 70, 3rd loop- 71 to 99%
- Oral Defense Manuscript Submission (3 Loops)
- Oral Defense Proper
- Oral Defense Manuscript Revisions

The team has the option to have the Mock Defense one (1) month before the Final Oral Defense. Mock defense is optional per the discretion of the group. A set of panel members will be invited to review and identify the requirements of the study and the application.

The panel will list all the requirements and recommendations for the team to accomplish for the Final Defense. Failure to comply any one of the recommendation will result in a failing verdict.

Mock Defense follows the same guidelines as the Final Oral Defense including documentation, presentation and payments.

V. Research / Capstone Project Team

The Capstone Project team is composed of at most five (2-5) members. The following are the roles that the proponents/researchers should play:

- **Project Manager** - The person with authority to manage a Research / Capstone Project. This includes leading the planning and the development of all Research / Capstone Project deliverables. The project manager is responsible for the budget, work plan and all Project Management Procedures (scope management, issues management, risk management, etc.).
- **Systems Analyst** – the person who checks that all parts of the system are coordinated.
- **Programmers** - The persons who design, write, and test computer programs.

- **QA Staff / Tester** - A person who ensures the quality of the software product and help find and eliminate any bugs. He determines the functionality of every aspect of a particular application.
- **Documenter /Technical Writer** - A person who writes the Research / Capstone Project study document, both the system and the Research / Capstone Project manuscript.

Duties and Responsibilities of the Proponents/Researchers

1) Keep informed of the Capstone Project Guidelines and Policies.
2) Keep informed of the schedule of Research / Capstone Project activities, required deliverables and deadlines posted by Adviser and Dean.
3) Submit on time all deliverables specified in this document as well as those to be specified by the Adviser and Dean.
4) Submit on time all requirements identified by the Capstone Project Oral Defense Panel during the Oral Defense.
5) Submit on time the requirements identified by the adviser throughout the duration of the Capstone Project.
6) Schedule regular meetings (at least once a month) with the Adviser throughout the duration of the Capstone Project. The meetings serve as a venue for the proponent to report the progress of their work, as well as raise any issues or concerns.

Regrouping Policy

To ensure project continuity and fairness among all Capstone Project participants, the following regrouping policy shall apply:

Eligibility for Regrouping

Regrouping shall be allowed only if fewer than three (3) members remain in a team transitioning from Capstone Project 1 (ITS321) to Capstone Project 2 (ITS410). In such cases, the group may be officially disbanded, and the remaining members may be reassigned to other existing groups, provided that the maximum allowable number of members per group is not exceeded.

Continuation by Remaining Member(s)

If the remaining one or two members choose to continue their original Research or Capstone Project, regrouping shall not apply, but must be done with the written consent of both the assigned Capstone Adviser and the Program Head. A revision of the project scope may be required to reflect the reduced manpower.

Group Change Policy

- Any request for a change of group membership or reassignment must be mutually agreed upon by all concerned students (both the releasing and receiving groups).
- A consultation must be conducted with the respective Capstone Advisers involved and the Program Head, who shall evaluate and approve the proposed reorganization.
- Final approval of group changes shall be documented in writing and submitted to the Capstone Coordinator for proper record-keeping.

Documentation and Approval

All regrouping requests must be supported by:

- A written letter of request, signed by all concerned students
- Endorsements from the involved Capstone Advisers
- Approval from the Program Head

Research / Capstone Project Adviser

The subject teacher is by default the adviser of all groups assigned to him or her.

Duties and Responsibilities as the Subject Teacher

1) Announce Research / Capstone Project areas (at the start of the each semester) to the students;
2) Conduct general meetings with the students to discuss the Capstone Project Guidelines, Policies and Deliverables, and to allow the students to raise and clarify issues;
3) Select a Proposal Hearing and Oral Defense panel for each team;
4) Schedule Research / Capstone Project activities, such as the deadlines of deliverables and Proposal Hearing and Oral Defense sessions.
5) Post schedules, Proposal Hearing and Oral Defense guidelines, requirements guidelines, and other announcements;
6) Submit collated grades of students under his/her section for that term.

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Appendix A.

Project Team Assignments Form

INFORMATION TECHNOLOGY AREA

Team Name	

Name and Signature	**Project Role**	**Email address / mobile#**

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Appendix B. Pre-Proposal Statement Template

INFORMATION TECHNOLOGY AREA

NOTE: 1 for each of the 3 to 5 titles chosen by the adviser (strictly word-processed)

Project Title:	
Proponents/Researchers:	1. 2. 3.
Objectives of the Study:	
Scope and Limitations of the Study:	
Software Specifications Hardware Specifications	

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Proponents/Researchers:

1)
2)
3)

Proposed Project Title:

Submitted by: _____ (Signature of Project Manager over printed name) Date: _____	**Noted:** _____ (Signature of Adviser over printed name) Date: _____
Recommending Approval: _____ (Signature of Program Head over printed name) Date: _____	**Approved:** _____ (Signature of the Dean over printed name) Date: _____

***Accomplish in 3 copies

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Appendix D. Progress Monitoring Report Form

(Progress Monitoring Report)

COURSE / SUBJECT : _____

PROPONENT/s/STUDENT/s:
_____
_____

TITLE: _____
_____
_____

EXPECTED COMPLETION DATE: _____
(Show as per Gantt Chart) _____
DATE: _____

(To be Filled up by the Student/s)	**(To be Filled up by the Adviser)**
Completed Chapters/ Parts	Comments and Suggestions
_____	_____
_____	_____
_____	_____
_____	_____
Chapter / Part in Progress	
_____	_____
_____	_____
_____	_____
Difficulties Encountered	
_____	_____
_____	_____
_____	_____
_____	_____
Signature of Student/s	Signature of Adviser over Printed Name

Appendix E. Research / Project Manuscript Outline

- **Title Page** (different for COMP401 and ITS410)
- **Acceptance Sheet**
- **Approval Sheet**
- **Dedication** (free form)
- **Acknowledgement**
- **Executive Summary** or **Abstract**
- **Table of Contents** (strictly use MS Word Table of Contents feature)
- **List of Figures and Tables**
- **Chapter I – Introduction** (consistent for both ITS 321 and ITS410)
    - Project Context
    - Purpose and Description of the Study
    - Objectives of the Study
    - Scope and Limitations of the Study
- **Chapter II – Technical Background** (consistent for both ITS 21 and ITS410)
    - Technicality of the project
        - Software Specification
        - Hardware Specification
        - Program Specification
        - Test Plan
    - Details of the Technologies to be used
- **Chapter III – Methodology** (initial for COMP401 and final for ITS410; choose only the parts that are applicable)
    - Environment (only for org-specific capstone project)
        - Locale
        - Population of the Study
        - Organizational Chart/Profile
    - Requirements Specifications
        - Schedule Feasibility
            - Gantt Chart
        - Economic Feasibility
            - Cost and Benefit Analysis
        - Requirements Modeling
        - Conceptual Framework
            - Input
            - Process
            - Output
        - Data and Process Modeling
        - Context Diagram

- Data Flow Diagram
- Program Flowchart (highlights only)
- Risk Assessment/Analysis

Technology Architecture

4.1 Technology Stack

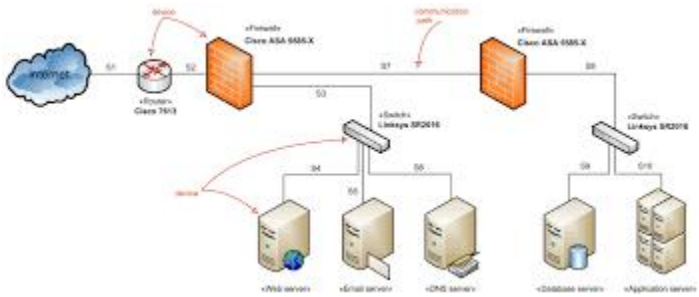
Present a structured list of all technologies used in the development of the project, divided into layers:

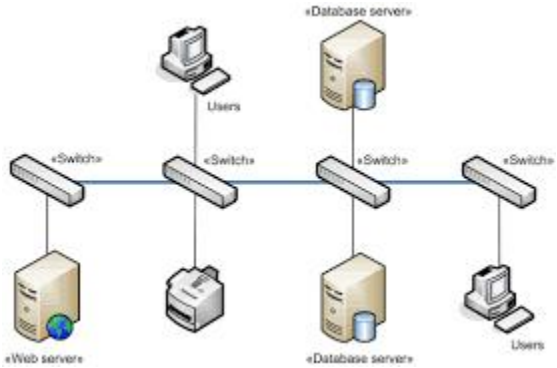
Layer	Technology Used
Front-end	e.g., HTML, CSS, JavaScript, React
Back-end	e.g., Node.js, PHP, Django
Database	e.g., MySQL, MongoDB
Hosting/Server	e.g., Firebase, AWS, Apache
Development Tools	e.g., VS Code, GitHub, Postman
APIs / Libraries	e.g., Bootstrap, Chart.js, Axios

Note: Customize based on your actual stack.

4.2 Network Architecture *(Required for web or LAN-based systems)*

Examples:





- Include a **network diagram** showing:
- How users/devices interact with the system
- Servers, routers, access points, and other hardware involved
- Data flow between client and server

Explain the network setup (e.g., Client-Server, Peer-to-Peer, Cloud-based)

- Design
  - Output and User-Interface Design
  - Forms
  - Reports
  - Data Design

  Entity Relationship Diagram (preferably done in MS Access [but MS Access is discouraged as DBMS])
  - Data Dictionary
- Testing
  - Functionality Testing
  - Compatibility Testing
  - Performance Testing
  - Acceptance Testing

PRESENTATION OF RESULTS

Proof of Functionality Testing

Presentation of Post Development Evaluation Results

- **Conclusions** (only in ITS410)
- **Recommendations** (only in ITS401)
- **BIBLIOGRAPHY**
- **APPENDICES** (initial for ITS321 and final for ITS410)
  - Letter to the Beneficiary
  - Relevant Source Code
  - Evaluation Tool
  - Technical Manual
  - Sample Input / Output / Reports
  - Users Guide
  - Other Relevant Documents
  - Working Title Form
  - Grammarian's Certification
  - Curriculum Vitae
- **GLOSSARY**(initial for COMP401 and final for ITS410)

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Appendix F. Research / Capstone Project Hearing Notice Form

RESEARCH / CAPSTONE PROJECT HEARING NOTICE

Date filed: _____

[ ] PROPOSAL
[ ] ORAL DEFENSE

Date of Defense: _____ Time: _____ Venue: _____

Research Title:

_____

_____

Proponent/s:

_____ _____ _____

_____ _____ _____

_____ _____ _____

C E R T I F I C A T I O N

The undersigned members comprising the panel for examination hereby agree to the schedule of hearing for the above project. [Please **PRINT NAME** and **SIGN**]

_____
RESEARCH ADVISER

_____
Program Head

_____
PANEL MEMBER 1

_____
PANEL MEMBER 2

_____
PANEL CHAIR

Appendix G. Grammarian's Certificate Template

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Date: _____

G R A M M A R I A N'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the proposed project study / research entitled

"_____" as against the set of structural rules that govern the composition of *sentences*, *phrases*, and *words* in the English language.

Signed:

_____
Grammarian

Conforme:

_____
Project Manager

Appendix H. Approval Sheet

La Consolacion College Bacolod
Galo corner Gatuslao Streets, Bacolod City, 6100 Philippines
Tel. Nos. +63(34) 4349661 to 64 | Fax +63(34)4335190
Website: www.lcc.edu.ph | Email: tph.bsit@lccbonline.edu.ph

APPROVAL FOR ORAL PRESENTATION

In partial fulfillment of the requirements for the Degree of Bachelor of Science in Information Technology, **Juan Dela Cruz,Juan Reyes and Juan Prinse**, who are hereby recommended for oral presentation have submitted this Capstone Project entitled **"Web Based Automation System".**

_____
Date

Adviser

PANEL OF EXAMINATIONS

Approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology by the Oral Examinations Committee for the Capstone Project with the grade of _____on _____.

Prof. Aileen Caday.
Chairman

Prof.Joseph B.Valenzuela
Member

Mr. Joe Cuenca De Guzman
Member

Accepted as partial Fulfillment of the requirements for the Degree of Bachelor of Science in Information Technology (BSIT).

Dean, SBIT

Appendix I. Title Page (Proposal)

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"AI-Based Facial Recognition Attendance Monitoring System for Educational Institutions"

(Must be inverted pyramid form, all caps)

A Proposal
presented to the Faculty of the
School of Business and Information Technology
La Consolacion College Bacolod

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By
Juan Dela Cruz
Peter Reyes
Luke Santos
Jude Paras
Paul Gomez

March, 2011
(the batch month and year)

Appendix J. Title Page (Final Oral Defense)

La Consolacion College Bacolod
Galo corner Gatuslao Streets, Bacolod City, 6100 Philippines
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"AI-Based Facial Recognition Attendance Monitoring System for Educational Institutions"

(Must be inverted pyramid form, all caps)

A Research / Capstone Project
Presented to the Faculty of the
School of Business and Information Technology
La Consolacion College Bacolod

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By
Juan Dela Cruz
Peter Reyes
Luke Santos
Jude Paras
Paul Gomez

March, 2011
(the batch month and year)

Appendix K. Title Page (Oral Defense)

La Consolacion College Bacolod
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INFORMATION TECHNOLOGY AREA

ACCEPTANCE SHEET

The Capstone Project concept entitled "**Web Based Automation System"** has been completed and submitted by the proponent.

Juan Dela Cruz
Peter Reyes
Luke Santos
Jude Paras

And is recommended for acceptance.

Adviser .

After a thorough review and evaluation of the Capstone Project criteria, the committee for the Capstone Project has accepted the title.

The acceptance is valid only to the information being presented.
Accepted this 1st Semester, School Year 2009 – 2010.

Dean, SBIT

Date of Acceptance: _____
Semester : 1st Semester
School Year : 2009-2010

GENERAL FORMAT OF THE CAPSTONE PROJECT

Below is the list of parts that must be followed. It is already arranged in the order that it should appear for easy reference. These parts will be discussed later.

1. **Blank Sheet**
   This serve as the flyleaf

2. **Approval Sheet**
   This page bears the name of the proponent, title of the thesis, signature of adviser, college dean, and members of oral defense. This page certifies that the thesis has been duly approved and must bear the date of approval with the mark received by the proponent. (See Appendix A)

3. **Title Page**
   Contains title of thesis, name of proponents, statement regarding the qualification for which the thesis is submitted. It also contains the name of the institution to the thesis is being submitted and the month and year of submission
   (See Appendix B)
4. **Acknowledgements**
   This section recognizes the persons/organizations that assisted the proponents in the completion of the thesis. Should be simple and tactful.

5. **Table of contents**
   A sequential listing of all major parts of a thesis with corresponding page numbers. Includes titles of chapters, sections, subsections, bibliography and appendices.

6. **List of Tables**
   Comprehensive listing of tables found in the body of the thesis with corresponding page numbers. Should be in sequence using Arabic numerals.
   Example, the third table in chapter 2 should be Table 2.3

7. **List of Figures**
   List of charts, graphs and other illustrations used in the thesis. . Should be in sequence using Arabic numerals. Example, the third figure in chapter 2 should be Figure 2.3

8. **Abstract**
   A presentation of thesis summary. It serves as a backgrounder for readers to have an overview of the study without having to read the entire thesis.

9. **Main Body**
   This is the main text of the thesis. It is divided into chapters and sub-topics. Starts with introduction and ends with conclusions and recommendations. This will be discussed separately.

10. **Bibliography**
    List of works cited, as well as works consulted in the construction of the thesis. For the format see Appendix C.

11. **Appendices**
    It should come after bibliography. Includes original data, preliminary test, tables and documents that the proponents feel are pertinent.

12. **Biographical Sketch**
    This shows the information about the proponent.

Body of the Project

1. Introduction

This chapter serves as a backgrounder for readers to have an overview of the study even without prior reference to other publications on the topic. This is the first chapter and must include the objectives, justification of the study and limitations set by the proponent. It is also the proper place to define any specialized terms used in the thesis.

2.Technical Background

This Chapter presents the hardware, software and programming environment that will be used** for the development of the project. Testing plans* and details of the technologies used** will also be presented.

3. Methodology

It is the ordered listing of steps and procedures used by the proponent/s. Methods used for gathering data, theoretical/conceptual framework, testing experiments as well as techniques employed in the analysis of data. The environment of the study, requirements specifications, designs and diagrams showing the totality of the system are also included.

4**. Presentation of Results**

This part presents the output of the system development and testing phase. It includes:

Screenshots or demos of the functional modules or features
User interface previews
Testing outcomes (unit testing, user acceptance testing, functionality tests)
Performance analysis based on usability, speed, accuracy, and reliability
Tables, graphs, and figures may be used to support the presentation and interpretation of data.

5. Conclusion and Recommendations

This chapter includes the results of the test done by the proponents on the system. Recommendations must be stated in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the thesis topic.

Capstone Project Physical Requirement

Paper

Use white letter size(8x11in). Orientation should be portrait style (can be read lengthwise). Text and figures are placed on only one side of the paper. The other side should be left blank.

Language

The official language to be used in writing a thesis is American language. Use only formal, objective expository English. Idioms, slang, subjective words and phrases, metaphors and undefined acronyms are not acceptable. All sentences must be complete with acceptable grammar and correct spelling.

Type of specifications

All submitted drafts of a thesis and other related official documents must be typewritten. Under no circumstances shall a hand-written draft and or form be accepted for submission. Font should be Arial twelve-point (12px) size for the text. Text color is black. Typing should always be neat, clean and uniform.

Margins

For the first page of any chapter, the top ,left and bottom margins should be 40mm wide.

For the rest of the pages, the left margin should be 40mm and the top and bottom should be 25mm.

The right margin is 25mm for all pages. Make sure texts*** are justified on both sides. Use five space indentation** for every paragraph.

Margin*** specifications are to facilitate binding and trimming.

Spacing

The body of the thesis should be typed double spaced

Use multi spacing for the approval sheet and title page for good visual effect.

Pagination

Pages should be numbered consecutively throughout the thesis, including pages of figures, tables appendices, and bibliography. Page 1 begins with the first page of the first chapter. Page numbers are not to be enclosed by parentheses, hyphens or other decorative devices.

Preliminary pages preceding the first chapter are numbered using small roman numerals(i,ii,iii…). The title page should not be numbered

Page numbers are placed at the center of the bottom of each page

Sample of a Bibliography

Example of a book	1. Patterson, F. and Linden, E. (1981) The Education Of Koko, New York:Holt,Rinehart
Example of a periodical	2. Otto, M.L. (1984), Child Abuse:Group Treatment Personnel and Guidance Journal, vol 62
Example of a magazine	3. Wang, .M. (1983). Talking with computers International Communications,pp 13-18
Example** of computer Software	4. Programmer's Aide [Computer Program],(1993) Los Angeles, CA. Software Creation Company
From Internet	5. gates, B.J. (1996, January), Talking Back to computers

BIBLIOGRAPHY

· CHED. (2015). *Policies, standards and guidelines for Bachelor of Science in Information Technology (CMO No. 25, Series of 2015)*. Commission on Higher Education. https://ched.gov.ph/cmo-25-s-2015/

· · Ala, J. S. (2020). *Capstone project in information technology education: Frameworks and best practices*. Journal of Computing and Information Technology, 28(4), 389–400.

· · Fabito, B. S., Celestino, M. A., & Macadaeg, C. E. (2021). *A framework for evaluating IT capstone projects in Philippine higher education institutions*. International Journal of Computing Sciences Research, 5, 1307–1320. https://doi.org/10.25147/ijcsr.2017.001.1.130

Fabito, B. S., Celestino, M. A., & Macadaeg, C. E. (2021). *A framework for evaluating IT capstone projects in Philippine higher education institutions*. *International Journal of Computing Sciences Research, 5*, 1307–1320. https://doi.org/10.25147/ijcsr.2017.001.1.130

Best Capstone Project Selection Guideline (Based on Panel Scores)

Award Basis: Highest Average Panel Score
The Best Capstone Project Award shall be granted to the group that achieves the highest overall weighted average grade as given collectively by the oral defense panel members, based on the official scoring rubric used during the final defense.

Evaluation Procedure:
Each panel member scores the project independently based on approved criteria (e.g., originality, functionality, documentation, etc.).
Scores from all panel members are collected and averaged per group.
The group with the highest final average score among all presenting teams shall be declared the Best Capstone Project.

In case of a tie:

The Capstone Coordinator and Program Head may initiate a recalibration discussion with the panel to determine if both or only one group will be awarded.
A supplemental criterion such as "User Impact" or "Deployment Readiness" may be considered to break the tie if pre-approved.

Eligibility Conditions:
The group must have passed all defense loops and submitted their final manuscript with incorporated revisions.
The group must have complied with all deadlines, requirements, and submission protocols.
The project must have been evaluated by a complete panel of three members.